



REVIEW ARTICLE

Designing electronic graphic symbol-based AAC systems: a scoping review. Part 1: system description

Kerstin M. Tönsing^a , Jessica Bartram^a, Refilwe E. Morwane^a  and Annalu Waller^b 

^aCentre for Augmentative and Alternative Communication, University of Pretoria, Pretoria, South Africa; ^bSchool of Science and Engineering, University of Dundee, Scotland, UK

ABSTRACT

Purpose: This is the first of two papers summarizing studies reporting on the design of electronic graphic symbol-based augmentative and alternative communication (AAC) systems, to determine the state of the field. The aim of this paper was to provide an overview of the general characteristics of the studies and to describe the features of the systems designed.

Methods: A scoping review was conducted. A multifaceted search resulted in the identification of 28 studies meeting the selection criteria. Data were extracted relating to four areas of interest, namely (1) the general characteristics of the studies, (2) features of the systems designed, (3) availability of the systems to the public, and (4) the design processes followed. In this paper, findings relating to the first three areas are presented.

Results: Most study authors were affiliated to fields of engineering and/or computer science and came from high-income countries. Most studies reported the design of AAC applications loaded onto mobile technology devices. Common system features included customizable vocabulary items, the inclusion of graphic symbols from both established AAC libraries and other sources, a dynamic grid display, and the inclusion of digital and/or synthetic speech output. Few systems were available to the public.

Conclusions: Limited justifications for many of the complex design decisions were provided in the studies, possibly due to limited involvement of rehabilitation professionals during the design process. Furthermore, few studies reported on the design of graphic symbol-based AAC systems specifically for middle- and low-income contexts and also for multilingual populations.

ARTICLE HISTORY

Received 14 June 2022
Revised 8 October 2022
Accepted 7 November 2022

KEYWORDS

Assistive technology;
augmentative and
alternative communication;
design; electronic AAC
systems; graphic symbols

► IMPLICATIONS FOR REHABILITATION

- Complex design decisions about electronic graphic symbol-based augmentative and alternative communication (AAC) systems should be made purposefully and with sufficient justification.
- Increased collaboration between designers and rehabilitation professionals during the design of electronic graphic symbol-based systems could improve the products.
- Design of AAC systems for populations residing in low and middle-income contexts and also for multilingual populations are urgently needed.

Introduction

Persons with complex communication needs (CCN) typically benefit from augmentative and alternative communication (AAC). Electronic graphic symbol-based (GS-based) AAC systems are a form of assistive technology employed to improve communication and participation of children and adults with CCN who are not literate [1–4]. Such systems consist of (a) graphic symbols¹ such as line drawings or photographs that encode words, phrases, or messages; and (b) an interface that displays these symbols and allows a person using the system to access them in some way, for example, by touching or selecting symbols using switches or a cursor. Many (but not all) electronic systems also provide voice or speech output, either in the form of synthetic speech or recorded (digital) speech [5–7]. Systems with the latter capability are also referred to as speech generating devices (SGDs). Designing GS-

based systems requires making a number of design decisions about the vocabulary to be made available, the types of graphic symbols used, the meaning units to be represented by these symbols, and the way the symbols are organized on the interface. In contrast to an orthography-based system, where the relationship between orthographic symbols and the linguistic unit is standardized according to the writing conventions of the language in question, graphic symbols are a non-standard and mostly incomplete representation of language. Designers are typically forced to engage in trade-offs between the breadth of messages and vocabulary that are being made available and the learning demands that extensive systems pose [6].

This paper is the first of two to describe the results of a scoping review of the literature that describes the design of electronic GS-based AAC systems. In this paper, we provide general information about the studies and then describe the systems that were

CONTACT Kerstin M. Tönsing  kerstin.tonsing@up.ac.za  Centre for Augmentative and Alternative Communication, University of Pretoria, Lynnwood Rd, Hatfield, Pretoria 0002, South Africa

 Supplemental data for this article can be accessed online at <https://doi.org/10.1080/17483107.2022.2147228>.

© 2022 Informa UK Limited, trading as Taylor & Francis Group

designed, with specific focus on the included vocabulary, graphic symbols, interface, access methods and the voice output possibilities. The availability of the systems to consumers is also described. In the second paper [8], the design process is described in more detail and the extent to which this process aligns with the principles of human-centred design is considered. The study is framed by a brief overview of GS-based systems, including their implementation, their components, and recent developments.

Implementation of GS-based systems

Graphic symbols are representations of language concepts and have been used in the field of AAC for many decades [9]. GS-based AAC systems may serve a variety of purposes, such as supporting comprehension and expression as well as behaviour and organization [2,10]. In young children with CCN whose receptive language is developing relatively age-appropriately, GS-based AAC systems have been used with the primary purpose of supporting their expressive communication [11]. There are numerous studies that attest to positive outcomes for individuals who received intervention using GS-based AAC systems, including electronic systems. Gains have been noted in the realm of semantics (e.g., receptive and expressive vocabulary), pragmatics (e.g., increased turn-taking), and morpho-syntax (e.g., increased length of utterance and morphological complexity) [12,13].

In spite of the positive outcomes reported in various studies, GS-based AAC systems are not always perceived as helpful, useful or usable by the intended beneficiaries, and rejection and abandonment have been reported in the literature [14]. Abandonment rates of AAC systems between 31% and 60% have been reported [15,16]. While a variety of complex factors contribute to the rejection and abandonment of GS-based AAC systems, various characteristics of the AAC system itself have been noted to play a role [17,18].

Components of GS-based systems and design considerations

As previously described, an electronic GS-based AAC system has an interface displaying graphic symbols that can be selected by the user. While some controversy exists regarding the exact definition of the term “graphic symbol” [cf. 7,19], we use it to denote pictures (including photographs) or line drawings used in the field of AAC to represent a word, concept, or phrase. Electronic systems may also have speech output capabilities by which the language unit is made audible once the symbol is selected. In order for the system to be usable and useful, aspects like portability, durability, and ease of access need to be considered in the design of the hardware, while aesthetics can enhance the desirability [20] and reduce stigma [21]. Access possibilities include those that require the user to directly select and activate a specific symbol, for example, using a finger to activate a symbol on a touchscreen, or using eye gaze to select a symbol with the help of an infrared video camera [5]. A user may also control a cursor with the help of a mouse or joystick to make direct selections. Scanning, in turn, does not require the motor control that direct selection requires. Symbol options are highlighted in turn by the device and the user activates a switch when the desired option is highlighted [5]. Pampoulou and Fuller [7] further distinguish between selection that is unaided (i.e., requires only a body part such as a finger) versus selection that is aided (i.e., requires a selection aid such as a mouse or a switch).

Various factors around the language representation system (i.e., the vocabulary items, the graphic symbol, as well as the layout and organization of these) as well as the presence and quality of voice output also need to be considered. Some of these factors are briefly described in the following paragraphs.

The preselection of vocabulary that will be made available to the user is one of the most challenging aspects in designing and customizing a GS-based AAC system [22]. As natural speakers can select and produce their vocabulary at will and on the spot, designers and practitioners often struggle to predict what vocabulary their clients may find useful in various situations [23,24]. While vocabulary should be immediately useful to enable communication and participation with maximal ease in multiple settings and contexts, it should also in many cases allow for increasingly sophisticated use of language, especially when the language skills of the user are expected to improve or develop [5]. These foci sometimes imply trade-offs, for example, between immediate usability (reduced learning demands) and long-term potential of the vocabulary to support novel utterance generation [5].

A number of different approaches and sources for vocabulary selection have been suggested and debated in the literature. The current state of knowledge in the field suggests that vocabulary needs to be selected from multiple sources [22]. These sources can include core vocabulary lists, developmental word lists (e.g., The MacArthur-Bates Communicative Development Inventories or CDI [25]), the suggestions from parents and other stakeholders, and environmental checklists [5,22,26]. While a preprogrammed set of words made available by the developer can be helpful as a starting point, a level of customization will always be required when adapting a system for a specific user. Just-in-time programming (i.e., programming new vocabulary on the spot) has been shown to be a helpful capability to allow instant and easy programming of relevant vocabulary as needed [27,28].

Regarding graphic symbols, various commercially and freely available graphic symbol libraries have been developed over the years, for example Picture Communication Symbols™ (PCS), Symbolstix™, Widgit™ and free symbol libraries such as Mulberry symbols, symbols developed by the Aragonese Centre for AAC (also called ARASAAC symbols) and Jellow. Most were originally designed to represent a specific language. For example, PCS, Widgit (developed from the Peabody Rebus Reading Program's symbols), and Blissymbols were originally designed to represent the English language [29,30]. As a result, additional development and adaptations are usually needed before such a symbol system can be used to represent another language, as vocabulary and grammatical elements differ between languages. For example, the agglutinative structure of the Turkish language makes many of the available symbol libraries unsuitable to adequately represent the language [31]. Furthermore, the visual nature of symbols lends itself to representing concepts in a way that is aligned to a specific geographical context and culture, and this may not translate to other geographical contexts and cultures [31–36]. Food, modes of transport, building styles, weather, clothing, religion and conventions around celebrations (e.g., weddings) are some of the obvious differences between cultures and geographical contexts. As a result, a snowflake may not be a logical symbol to represent the concept ‘cold’ in some countries or regions – to give but one example.

Graphic symbol collections differ on a number of variables, for example, complexity, distinctiveness, figure-ground differential, iconicity, and vocabulary size and representational range [7,19]. These variables influence how easily the symbols are recognized and/or learnt, and also to what extent they allow the user to

generate novel utterances [37]. Pampoulou and Fuller [37] suggest that AAC symbols should primarily be classified along a continuum of linguistic properties. According to these authors, non-linguistic symbols like photographs and clip art images do not possess an internal logic by which new symbols can be created. These collections are also used idiosyncratically (i.e., compiled for a specific person) and the number of vocabulary items represented is typically limited. These factors limited the user's ability to generate a wide range of novel messages. Many symbol collections consisting of line drawings (e.g., PCS and Widgit) are classified as prelinguistic. These symbol collections possess a degree of internal logic, and often represent a wide range of concepts and different parts of speech. They allow the user to generate a variety of utterances and meanings, although they do not give access to the unlimited generativity that a truly linguistic system (such as orthography) offers. The linguistic properties of the graphic symbols designed or incorporated into the AAC system need to be carefully considered to ensure maximal communication autonomy and (specifically in children) language development [37].

Regarding vocabulary organization (i.e., lay-out), visual scene displays (i.e., a highly contextual picture or photograph of a scene with embedded hotspots that contain pre-programmed messages) have been shown to facilitate message retrieval and communication turns in young children as well as adults with chronic severe aphasia [38,39]. Visual scene displays have been described as establishing a contextualized, personally meaningful and event-based platform for the exchange of messages that is of benefit to beginning communicators. Grid-based displays are more commonly used for individuals at a higher level of language development than visual scene displays [40]. Theoretically, grid displays do not "lock" a vocabulary item into a specific context [41] and may therefore encourage more generalized use. According to Beukelman and Light [5], vocabulary items on grid displays have been arranged (a) by category (also referred to a taxonomic organization, e.g., people, food, toys, etc.); (b) by schema (e.g., by event, activity or setting); (c) by part of speech (e.g., nouns, verbs, etc.) and/or syntactic category (e.g., subject-verb-object); or (d) alphabetically. In addition, it has been suggested that core vocabulary should be displayed in a prominent location on an AAC system [26]. Often, more than one organizational approach is combined within one system.

Research evidence supporting the choice of a particular organizational approach over another is scarce. Although Fallon et al. [42] found that typically developing children aged 4–5 years tend to organize words by event schema, Light et al. [43] found that they learnt to locate vocabulary items on both schematic and taxonomic grid displays with equal accuracy. Alphabetical organizations rely on phonemic knowledge and are therefore typically not appropriate to preliterate individuals. Thistle and Wilkinson [40] surveyed 112 SLPs on their decision making around AAC display design. They found that grid-based designs were used most often with a strong focus on the consistency of the vocabulary display to capitalize on motor planning. Motor planning and display consistency can reduce learning demands including reducing the overload of working memory [44–46]. It has been noted that schematically organized vocabulary often leads to duplication of vocabulary items in various locations [43].

The voice output of electronic AAC systems can be digital (recorded), synthetic (i.e., computer-generated, also often referred to as text-to-speech technology), or hybrid (combining both synthetic and digitized speech) [7]. Digital speech output requires that a speaking person use his or her voice to record every word

or phrase that is to be made available on the system. Preferably, all recordings should be made by the same person in order to give some uniformity, and the voice should be well-matched in terms of age and gender [47]. If the person recording messages is a familiar partner, this may lead to some confusion as the person using the device "speaks" with the partner's voice. When recordings are made by someone who is not a familiar partner, adding new recordings may be difficult, as this person may not always be available. Programming synthetic speech into a device is not dependent on a specific person, as the message to be spoken by the system is merely typed into the device. Innovations in the methods used to produce synthetic speech have led to a reduction in cost and significant improvements in quality over the last 20–40 years [47]. However, text-to-speech technology has not yet been developed for all languages, partly due to resource constraints [48,49]. Also, for some languages, only adult voices and/or only voices of a particular gender may be available [48]. Tone of voice remains problematic in all synthetic speech, and the person using the system typically cannot choose the emotion or emphasis that is to be conveyed by the voice [47].

GS-based systems: recent developments

The mobile technology revolution has led to a proliferation of electronic AAC applications and software that can be loaded onto tablets, laptops and mobile phones [50]. Such non-dedicated AAC systems have become more accessible and affordable to many individuals who use AAC when compared to older, dedicated devices [51]. Additional features such as Internet access, texting, and social media allow for a wider range of communication opportunities [50]. The use of mainstream devices has reduced stigmatization of this form of assistive technology [51–53]. In addition, advances in computing have made a number of additional AAC system functionalities and features possible, such as animated symbols, symbol prediction and pop-up windows with grammar and morphology options [54].

The reduced cost and simplification of the design process has democratized the development and access to electronic AAC systems in many ways. Individuals who may not have a high level of expertise in AAC or rehabilitation engineering can now develop AAC systems, and the mainstream availability and relative affordability of both hardware and software has reduced the reliance on healthcare professionals to drive and oversee the acquisition of electronic AAC systems [50,55]. This development has also opened up possibilities for AAC system development in less resourced contexts – both contexts with fiscal constraints and also those with AAC knowledge and expertise constraints. There are indications that AAC systems are being increasingly developed for people from a variety of linguistically and culturally diverse and previously underserved contexts [56].

At the same time, a solid research base to elucidate the interaction between persons with CCN, the AAC system, the partner and the environment is still lacking, and AAC system design remains technology driven rather than research based [57]. As with previously designed AAC systems, AAC applications may improve some aspects of communication for some clients and their partners, but the usability, usefulness and the user experience they offer may not be optimized. When the emphasis remains on the application and construction of the technology, the psychological and sociological impact of the design, and the goodness of fit to the needs, preferences and contexts of a specific population are often ignored [58].

The proliferation of non-dedicated electronic AAC systems and burgeoning developments to design such systems for non-English and preciously underserved contexts suggests that a review of design processes may be a timely endeavour. In this paper, we will provide general backgrounds of the studies. We will also describe the systems that were designed, with specific focus on the included vocabulary, graphic symbols, interface and access possibilities and the voice output possibilities. The availability of the systems to consumers is also considered. In the companion paper, we describe the design approaches and processes used, and the extent to which human-centred design principles were incorporated in the design of new AAC systems. Because design considerations for GS-based systems may differ from those for orthography-based systems, we decided to limit this review to the former type of system.

Method

A scoping review was conducted to identify and synthesize the literature that documents the design of electronic GS-based AAC systems. Scoping reviews summarize the literature more broadly without focusing on a very specific and narrow research question [58]. The scoping review procedure was guided by the methodological framework developed by Arksey and O'Malley [59] enhanced by Levac et al. [60], and the PRISMA extension for scoping reviews (PRISMA-ScR) [61]. However, the optional sixth stage (consultation with stakeholders) of this methodology was not conducted. A review protocol was developed prior to conducting the review.

Search strategy

A multifaceted search strategy was developed by the authors in consultation with a subject librarian. The search aimed to cover literature from all relevant fields, including rehabilitation, education, linguistics, engineering, and computer science. The search methods included the following:

- Electronic database searches (Advanced Technologies & Aerospace Collection, Academic Search Complete, Computer Science Database, Cumulative nursing and Allied Health Literature [CINAHL], Education Collection, Educational Resources Information Centre [ERIC], Health Source: Nursing/Academic Edition, Humanities Index, Humanities Source, IEEE Xplore Digital Library, Library & Information Science Collection, Linguistics Collection, Linguistics and Language Behavior Abstracts [LLBA], MEDLINE, ProQuest Dissertations & Theses Global, PsycARTICLES, PsycINFO, Science Database, Social Science Database, Technology Collection, Springerlink, Taylor & Francis [journals], Wiley Online Library, Web of Science).
- Search using the Google Scholar search engine.
- Hand search of reference list of identified articles.

The electronic database search was conducted on 4 November 2020. A Really Simple Syndication (RSS) feed and search alerts were set up on all databases. Results from these feeds were collected until January 2021.

Pilot searches were conducted to assess the feasibility of the search terms. After refining and adjusting search terms, three search strings were compared between various databases to determine which search string provided relevant results. Relevant databases were then searched using the final Boolean search string: ("augmentative and alternative communication" OR AAC

OR "alternative communication") AND ("design process" OR "development process"). Limiters were set on language (English) and date (2000 – present). The databases, search string and yields are provided in Table 1.

Selection of studies

Studies were selected according to a list of inclusion and exclusion criteria, that were refined and finalized while piloting these criteria on pilot search results [60]. Table 2 provides the final inclusion and exclusion criteria that was used to select studies for this review. Screening was completed using Covidence, a web-based review platform [62]. Each record was independently screened by the first author and either the second or third author and evaluated for in- and exclusion based on the criteria established. Screening was conducted in two rounds – first on title and abstract level, and then on full text level. The percentage agreement after the independent screening was calculated by dividing agreements by the sum of agreements and disagreements and multiplying by 100. The agreement between the authors was 97.3% for the title and abstract screening, and 83.4% for full text screening. Any disagreements between reviewers were discussed, and consensus was reached as to whether to include or exclude the record.

Data extraction

A data extraction form was developed on Microsoft Excel in accordance with the sub-aims of the study. Data were extracted in accordance with the following categories (a) general characteristics of the studies (authors, date of publication, aim, country in which the development took place) (b) description of the product/prototype that was designed), (c) design approaches used, and (d) human-centred design principles observed in the study. In this paper, the results of data extracted according to categories (a) and (b) are reported. Data were extracted independently by the second author and one of two research assistants, who were qualified speech-language therapists with Master's degrees in AAC. Prior to the data extraction process, the assistants were provided with a verbal explanation of the information required for data extraction, as well as additional written definitions to ensure appropriate understanding of the data to be extracted. The data extracted by the second author and research assistants were compared and the initial agreement on data extraction before reaching a consensus was 89%. The author and the two respective research assistants discussed differences and reached consensus on most aspects, resulting in an agreement of 99.5%. The first author acted as arbitrator in the remaining 0.5% of disagreements. In preparation of this paper, the first author additionally extracted information pertaining to the system designed. The second author independently extracted the same data and an agreement of 99.8% was found. Consensus was reached on disagreements.

Results

Search results

The results of the search are documented using Tricco et al.'s [61] PRISMA flow diagram structure, as seen in Figure 1. A total of 3628 records were identified through electronic database searches, and an additional 97 records were identified via other sources (hand search and Google scholar search). Once duplicates were removed, 3256 records remained. Of these, 3155 were excluded during title and abstract screening. A total of 101 records were then screened on full text level, and 73 were

Table 1. Search strategy and number of citations generated per database.

Database	Search strategy	Initial yield	RSS Feed	Total (minus duplicates)
Scopus	ALL ("augmentative and alternative communication" OR AAC OR "alternative communication") AND ("design process" OR "development process")	164	3	146
EBSCOhost (Academic Search Complete, CINAHL, ERIC, Health Source: Nursing/Academic Edition, Humanities Source, MEDLINE, PsycARTICLES, PsycINFO)	TX ("augmentative and alternative communication" OR AAC OR "alternative communication") AND TX ("design process" OR "development process")	206	29	91
ProQuest (Education Collection, Humanities Index, Library & Information Science Collection, Linguistics Collection, ProQuest Dissertations & Theses Global, Science Database, Social Science Database, Technology Collection)	("augmentative and alternative communication" OR AAC OR "alternative communication") AND ("design process" OR "development process")	271	24	216
IEEE Xplore Digital Library	("augmentative and alternative communication" OR AAC OR "alternative communication") AND ("design process" OR "development process")	1065	7	1065
Taylor & Francis (journals)	("augmentative and alternative communication" OR AAC OR "alternative communication") AND ("design process" OR "development process")	267	2	215
Wiley Online Library	("augmentative and alternative communication" OR AAC OR "alternative communication") AND ("design process" OR "development process")	247	10	207
Computer Science Database	("augmentative and alternative communication" OR AAC OR "alternative communication") AND ("design process" OR "development process")	676	–	670
Linguistics and Language Behavior Abstracts (LLBA)	("augmentative and alternative communication" OR AAC OR "alternative communication") AND ("design process" OR "development process")	19	–	19
Advanced Technologies & Aerospace Collection	("augmentative and alternative communication" OR AAC OR "alternative communication") AND ("design process" OR "development process")	127	–	97
Springerlink	("augmentative and alternative communication" OR AAC OR "alternative communication") AND ("design process" OR "development process")	560	–	488
Web of science	("augmentative and alternative communication" OR AAC OR "alternative communication") AND ("design process" OR "development process")	26	–	22

excluded. A total of 28 studies remained for inclusion in the review. An overview of the studies included is provide in [Supplementary Appendix A](#).

General characteristics

[Supplementary Appendix A](#) provides some general characteristics of the included studies. It is noteworthy that most articles ($n=20$) were published between 2011 and 2020, while three were published between 2006 and 2010, and two between 2000 and 2005. Most studies included researchers from various disciplines/institutions. Most of the authors were affiliated to departments or schools of computer science, informatics/bioinformatics, and engineering/bioengineering ($n=17$). Fewer studies had authors affiliated to departments of health and rehabilitation ($n=8$) and education/special education ($n=5$).

Studies took place in various locations. A total of 12 studies took place in a European country such as Italy, Croatia, Portugal, Spain, the United Kingdom, and the Netherlands. Fewer studies were based in Asian countries ($n=7$) such as Japan, China, and India. The remaining studies were conducted in North American ($n=5$) and South American ($n=4$) countries. No studies were conducted in any African country or in Australia. A total of 22 studies were conducted in high income countries, whereas five studies were conducted in upper middle income countries and one study in a lower middle income country [63].

AAC systems designed

Most of the systems designed ($n=26$) were AAC applications designed to be uploaded onto commercially available hardware. In 18 studies, the hardware mentioned comprised of a form of mobile technology such as a smartphone, tablet, or personal digital assistant (PDA). Two studies mentioned computers without further specification, while one study mentioned a desktop computer. Five studies that described an AAC application did not specify the hardware on which it was loaded. In 12 studies, specific operating systems were mentioned, including a combination of iOS and Android ($n=4$), Android only ($n=4$) and iOS only ($n=3$). The Windows operating system was only mentioned once.

In two older studies, dedicated AAC systems were described. Allen [21] designed the Portland Communication Aid, consisting of a book with keyboard, waist pack, and mobile speaker unit. Hill et al. [64] designed a Mandarin software application for a 128-location speech generating device.

Systems were designed to give access to different languages. English was reported most frequently ($n=7$). Spanish ($n=3$) and Portuguese ($n=3$) were the next most common languages. Systems to give access to expression in Japanese ($n=2$), Chinese ($n=2$), and Arabic ($n=2$) were also reported. Only one study each reported on systems that gave access to Dutch ($n=1$), Mandarin ($n=1$), Hindi and Bengali ($n=1$), and Croatian ($n=1$) as languages used by the target population. In six studies, the language associated with the system was not mentioned. Only one

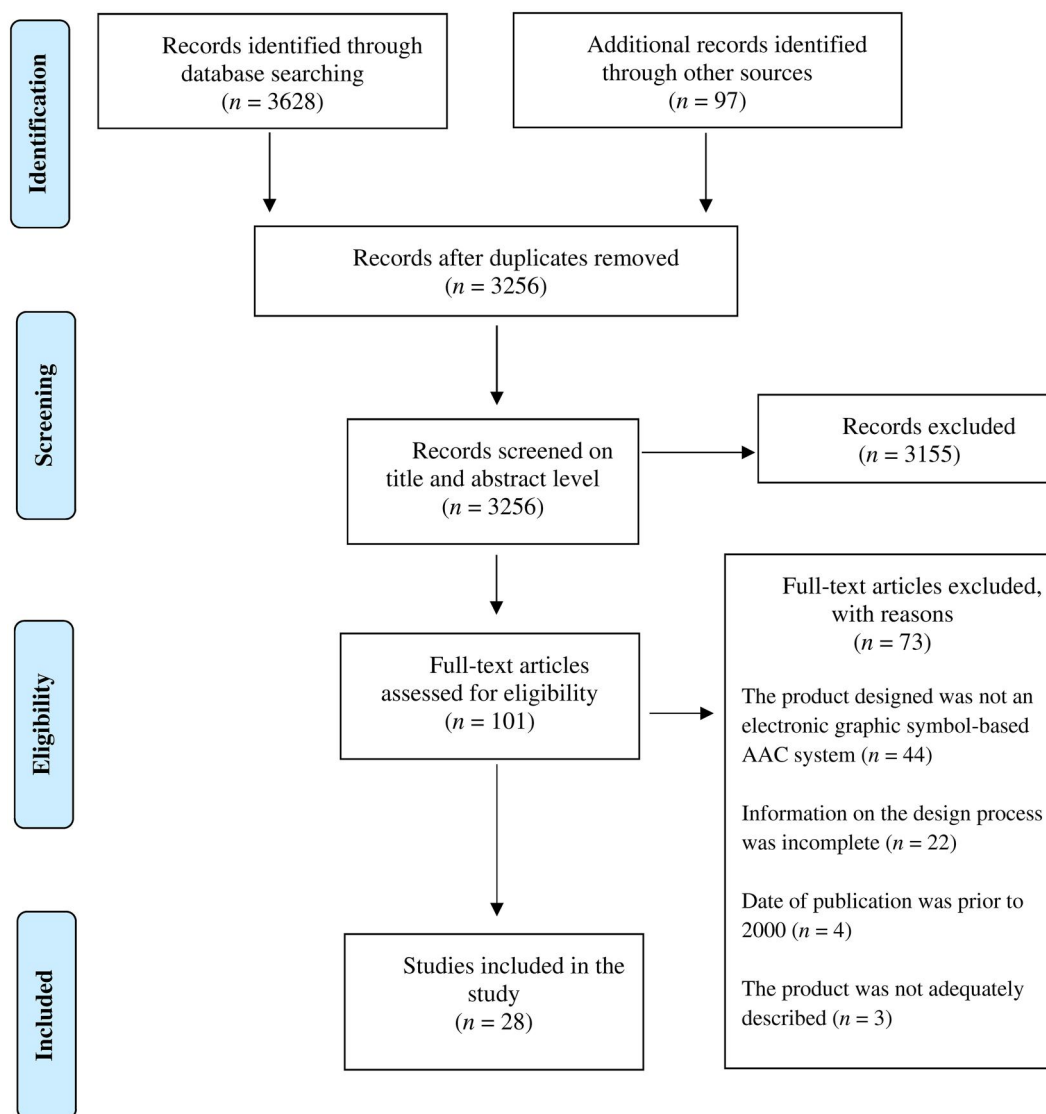


Figure 1. PRISMA flow diagram of the study identification and selection process.

system was designed to give access to more than one language, namely Hindi and Bengali [65].

The predetermined vocabulary included within each system was often not explicitly discussed, and hardly any information about the type and number of words/messages and/or parts of speech was provided. However, the incorporation of core words was mentioned by Hill et al. [64]. Al Arifi et al. [66] mentioned including the message 'I want' and customisable request items. Cheung [67] reported that vocabulary was categorized by parts of speech. Karita [68] and Rodríguez-Sedano et al. [69] seemed to base their vocabulary on that of other AAC systems. In contrast, the systems' capability to allow customization of vocabulary (i.e., the programming of personalized words or messages by users or their partners) was reported in 25 studies. Just-in-time (JIT) programming was mentioned in one study [70]. Karita [68] described a system that automatically changed the available vocabulary based on GPS input. Additional language features such as word or sentence prediction ($n = 3$) and access to morphology or syntax ($n = 3$) were also reported.

In various studies ($n = 10$), authors described including GS from one or more commercially or freely available AAC symbol

library such as PCS ($n = 4$) and ARASAAC symbols ($n = 5$). Other symbol libraries that were each mentioned once each were Widgit symbols, Sclera, and Mulberry. In nine studies, authors described drawings and pictures from other sources, including GS developed for previous AAC systems ($n = 3$; i.e., Lingraphica symbols, symbols developed by the Indian Institute of Cerebral palsy, and icons developed for a European AAC system). Images from clipart, web images in general, and images created specifically for the system (e.g., using Corel Draw X7) were also mentioned ($n = 3$). The option for the system user to create drawings or sketches within the AAC system (e.g., on a drawing pad) while interacting with others in order to convey their message was mentioned twice – in both cases as part of systems created for persons with aphasia [71,72]. A total of 19 designed systems incorporated photographs – these included photographs of objects, events or people taken by the individual using the device or retrieved from the Internet.

In many of the studies ($n = 18$), authors reported using a dynamic grid display, where selection options were presented in a grid array. Four systems were developed to facilitate storytelling and narration, and therefore made use of images (pictures or

Table 2. Criteria for inclusion of studies.

Aspect	Inclusion	Exclusion
Population	Persons with CCN. This could include individuals with the following diagnoses: physical disabilities such as cerebral palsy (CP), intellectual and/or developmental disabilities such as autism spectrum disorder (ASD) and Down's Syndrome, as well as acquired disorders such as stroke and traumatic brain injury (TBI) [28,29].	Children/adults without disability. Persons with learning impairments such as dyslexia.
Issue	Describes the design and/or development process of an electronic GS-based AAC system, including the input obtained and an evaluation of the product.	Describes only customization, testing, or implementation of a GS-based AAC system. Information regarding the design process is not adequately described.
Outcome	The outcome of the design process is an electronic graphic-symbol-based AAC system, defined as any AAC system or AAC application that requires a source of electricity (e.g., battery) to function, and which consists of graphic representations that are organized and displayed on an interface. An individual using the system can access these representations in some way, for example, by touching or selecting symbols using switches or a cursor (Beukelman & Light, 2020). Articles were included even if they reported on the design of multiple products; however, only information pertaining to the design of the GS-based AAC system was considered.	The outcome is any other AAC systems, such as an electronic text-based system, or a non-electronic system. The outcome is only one component of an electronic graphic-symbol-based AAC system (e.g., symbol library, synthetic voice). The outcome (product) is not adequately described.

photographs) presenting scenes, that were presented in a sequence, for example, by scrolling. Of these, two systems were designed for adults with aphasia [71,73]. In the remaining studies, the lay-out was not clearly described. Visual scenes were not explicitly mentioned.

The size of the symbols and colour coding was only reported in eight and seven studies respectively. Colour coding techniques included the use of the Fitzgerald key to categorize vocabulary, as seen in de Oliveira et al.'s [74] study. Hervás et al. [75] included a similar strategy and coloured coded vocabulary categories that correspond to the respective "part-of-speech tags" (p. 5653). The authors of five studies reported on additional display features such as the use of bright colours to maintain children's attention [76], categorizing pictures based on the time they were taken [71] or changing the graphic symbols based on location [68].

Synthetic or digital voice output as part of the GS-based AAC system were reported by the authors of nine and seven studies respectively. AAC systems that include both synthetic and digital voice output were reported in eight studies. In four studies the type of voice output was not specified. Regarding access methods, the authors of most studies ($n=21$) reported using touch screens as a way in which users could access specific graphic symbols on an AAC system. Other access features such as a keyboard ($n=4$), a mouse ($n=4$), switches ($n=3$), eye gaze ($n=1$), as well as a keypad for scrolling ($n=1$) were also mentioned.

Evaluation and availability of the product

The designed systems were not always final products available to the public, but also included prototypes. Of the products developed, 14 were reported to meet the requirements set out by the researchers, whereas 13 products partially met the target requirements. There was no indication from the studies that any of the products did not meet any of the target requirements. The need to redesign or adjust the products/prototypes designed was reported in eight studies. For example, Karita [68] suggested improvements regarding the user interface and the product

usability and Williams et al. [77] reported on refining the input mechanism and vocabulary structure of the system designed.

The availability of the products and/or prototypes designed was not commonly reported. The authors of only eight studies discussed the availability of the products designed, of which four are freely available and two are commercially available for purchase and thus are accessible to the public. The two remaining studies were not as specific. For instance, da Silva et al. [78] reported that their product was available but did not specify whether the product was freely or commercially available, whereas Bhattacharya and Basu [65] stated that the product was deployed at a number of institutions but did not specify if this product was also available for the public.

Discussion

This review underscores the proliferation of GS-based AAC systems designed as applications for mobile technology. The availability, relative affordability, and acceptability of this form of technology make it an attractive choice as a hardware platform for AAC systems [50,52]. It is furthermore noteworthy that although most studies were conducted in high income countries, six were conducted in upper middle or lower middle income countries. This may be another suggestion that non-dedicated AAC technology may be more affordable and obtainable in contexts that are less resourced.

From the prolific involvement of persons affiliated to departments of engineering and computer science, it is clear that these disciplines typically drive the design of GS-based AAC systems. Involvement of rehabilitation specialists in design teams was notably less, in spite of the important role such professionals could play in ensuring that user needs and abilities are considered more broadly (i.e., not only based on a small group of cases). In this way, poor feature matching and system abandonment can be avoided [50]. Some commercially available GS-based AAC systems currently include practitioner-designed features, such as vocabulary sets designed on the Gateway language strategy

conceptualized by Bruno [79], and page sets based on the Pragmatic Organization Dynamic Display (PODD) strategy by Porter [80].

While some of the features of designed systems clearly exploited the possibilities of the mobile technology (e.g., the use of the inbuilt camera to take photographs and incorporate these into the AAC system, as well as the use of the GPS system to automatically detect user location and change overlay based on location), the use of more traditional features such as grid displays was still evident in many studies. According to Judge et al. [81] and Light, McNaughton et al. [82], many electronic GS-based systems are based on older paper-based systems, which typically consisted of grid-based displays. It was interesting to note that VSDs were not mentioned, even though these types of displays have been receiving some attention in the recent literature. In some cases, this may suggest that designers take over standard practices without consideration of the latest research evidence [43,50]. Similarly, the potential of the newest technology to simplify certain operational processes in using an AAC system could lead to better system uptake [83].

Limited consideration also seemed to be given to the complex decisions around vocabulary selection and organization of the symbols, possibly reflecting a lack of understanding on the part of the designers. The availability and type of vocabulary is a factor that requires much consideration [52]. This review found that few studies considered the type of vocabulary in terms of the specific words used or grammatical structures (e.g., parts of speech) pre-programmed into the system. However, a positive outcome from this review was that many systems were designed to easily customize vocabulary, which included strategies such as JIT programming.

It is interesting to note that less than half of the developed systems included a specific predeveloped AAC symbol library. Amongst the systems where this was included, the popularity of the ARASAAC symbol library was notable – although it contained (at the time of writing) a smaller number of GS and has been more recently established than older libraries such as PCS and Widgit, the possibility of freely using it under a creative commons license seems to have made it an attractive option. Reason for minimal use of established symbol libraries may have included possible licensing questions and fees, and a lack of appropriateness of the images for the culture, language and context of the intended system users. However, it was nevertheless noteworthy that none of the remaining studies reported any details about the development of an extensive language and culture-appropriate symbol library. Instead, the incorporation of personalized images was often reported – this included generating one's own images using drawing tools, sourcing images from the web and taking photographs with the built-in camera of the device. Personalized AAC systems can better accommodate and reflect each individual's desires, preferences, culture and linguistic background [84–86]. At the same time, abstract vocabulary is usually difficult to represent with the typically more iconic images found on the web or represented by photographs (e.g., it may be difficult to take a photograph that represents “have” or “year”). Established AAC symbol libraries with tens of thousands of images representing different words may allow a somewhat more unified approach to represent a larger vocabulary.

Colour-coding GS was reported in some studies. This is a promising result as the colour of symbols and background plays a role in perceptual processes and can contribute to the ease of discriminating, recognizing and memorizing GS within a system, as well as improve reaction times when locating targeted symbols

[87]. This is important as it shows some form of consideration of the individual's visual and visual processing abilities and an effort to reduce perceptual and cognitive demands.

A total of 17 systems incorporated synthetic speech. This finding reiterates the importance of this technology for electronic AAC systems [5]. It was interesting to note though that, despite the advances in synthetic speech technology, seven systems only incorporated digital speech. These systems were developed to give access to English, Spanish, Japanese and Croatian. Synthetic speech technology exists for all of these languages, although it may not have been developed for the particular operating system at the time the study was conducted. Alternatively, its incorporation may have been too costly or time consuming. The more natural intonation that is possible using digital speech may have been another reason for choosing it. A combination of digital and synthetic speech as reported for eight systems allows for the flexibility to choose which option to use.

Most studies reported a touch screen as an access feature (e.g., the individual can select the relevant GS by touching the screen of the device). A touch screen is a common feature of smartphones and/or tablets. However, few studies reported on additional access features such as eye gaze, switches, and using a mouse to select GS. In light of the possible physical (motor control), visual, and cognitive limitations of the population for whom many of these GS-based AAC systems were designed, one may have expected to find more studies considering these additional access features [5,88].

It is clear from the limited number of available GS-based AAC systems reported within this review that many studies/designs are still ongoing. In addition, despite many systems meeting or partially meeting the requirements set out in the beginning of the design process, numerous studies have suggested recommendations for further development, as well as further research to test and/or improve the designed system. This illustrates the cyclical relationship between research and development that is typical of a design process [89,90].

Limitations

Although this review followed a rigorous search protocol, studies may have been excluded due to various aspects such as the limitation of only sourcing English articles, which may have influenced the number of citations yielded. Additionally, as this review addresses a large, multidisciplinary field, it may be possible that relevant citations may have been reported on databases that were not considered and as a result may have been excluded from this review. Possible exclusion of citations based on the identified search terms must be also acknowledged.

Moreover, considering that design processes are multifaceted, such processes may have been reported in numerous papers. Therefore, some relevant design studies may have been excluded from the review as the individual papers reporting on the process may not have met all the inclusion criteria. For example, a particular record may only have reported on designing one aspect of a GS-based AAC system (e.g., selecting appropriate vocabulary, or designing an appropriate symbol library).

The exclusion of grey literature in the review is another important limitation. Companies developing AAC systems may not have the access, funding, or interest to publish their development processes in peer-reviewed journals as their main mandate is not to produce and disseminate research findings. As a result, the design processes they use would not be documented in peer-reviewed articles.

As this was a scoping review, quality appraisal of included studies was not conducted and thus it is possible that the quality of studies may have varied. In addition, as is typical of a scoping review, many aspects of the included studies were described without going into much depth on any specific aspect. Instead of merely documenting the characteristics of the designed systems, for example, one could consider to what extent these characteristics matched the characteristics of the population, based on previous research on feature matching.

Recommendations for further studies and design projects

The findings of the review suggest that the specific features of the GS-based AAC could be better described in design studies, and that designers may be encouraged to pertinently consider and justify the decisions they make about the vocabulary, the layout, the graphic symbols included, the access methods, and the voice output included as part of the system. The findings also suggest that there is a greater need for collaboration between designers and engineers and rehabilitation professionals. Rehabilitation professionals, especially those working in the field of AAC, should be part of the design team as active co-designers. There are a multitude of reasons for this but for the most part, they can guide methods to involve users and stakeholders, suggest relevant outcomes to be evaluated, and may be more skilled at designing research to evaluate performance and obtain social validation. A review of grey literature (e.g., blogs and company websites) as well as interviews with developers and surveys of the industry could further be useful to understand to what extent rehabilitation professionals are already involved in system design and development.

It is clear that most of the design and development work in the field is still conducted in high-income countries. The emerging work in middle- and low-income countries identified in this review is encouraging, but more needs to be done in this regard. More than 80% of the world's population resides in low- and middle-income countries, where risk for and prevalence of disability is significantly higher [91]. High adoption rates of mobile technologies in low- and middle-income regions represent an opportunity to facilitate the roll-out of mobile technology-based AAC solutions [92,93]. GS-based AAC systems that are specific and appropriate to the cultures, languages, and contexts of the populations residing in low- and middle-income countries are urgently needed [94]. Similarly, systems that give access to multiple languages need to be designed. Although over half of the world's population is multilingual [95], most AAC systems are designed to only give access to one language.

Conclusion

The identification of 28 studies reporting on the design of a GS-based system is a positive finding, attesting to continued development of this form of assistive technology for the benefit of persons with CCN. The diversity of countries where these design projects were conducted and the diversity of languages for which systems were designed is another positive finding, suggesting that the availability of linguistically and contextually relevant electronic GS-based AAC systems is expanding to previously underserved populations. The proliferation of mobile technology in both high and low resource contexts has created a platform for making such systems more easily available and more affordable. However, it is clear that developments are still primarily taking place in high-resource contexts.

The findings further highlight that GS-based system design remains complex, and that designers are required to take a multitude of factors into consideration. In this paper, we have only focused on the system itself. Even here, many aspects require consideration, such as the vocabulary included, the GS used, the display, the voice output, and access. From the results of this review, it seems that designers do not always report in detail on the way each of these factors is addressed in the design process, and limited justifications are provided for specific design decisions. It was interesting to note that many systems designed seemed to require a fair degree of customization and programming from the support team of the person using the system, such as, for example, programming the vocabulary to be included, sourcing images from the web or taking personal photographs, and recording the spoken messages associated with the symbol. While such customization possibilities are positive to ensure an individual good fit between the system and the person using it, this places high demands on the support team. For persons with more advanced language skills, a more elaborate system with a larger preprogrammed vocabulary may be desirable.

One has to acknowledge that research to guide GS-based AAC system design decisions is limited [57], and often somewhat ambiguous. While specific design decisions may facilitate one aspect of system use (e.g., the use of VSDs to enhance immediate and contextualized use of a system), the same design decisions may not optimize other aspects of use (e.g., VSDs may not be ideal for generalized and flexible use of vocabulary). Furthermore, research on the effect of different system features on system use may not be readily available and accessible to computer scientists and engineers, as it is published primarily in the rehabilitation, psychology and education literature. There seems to be a measure of disconnect between the publications that focus on system design and those that focus on system use.

One interesting finding was that despite research attesting to the usefulness of many popular AAC systems sold by the prominent technology companies, information on the design of such systems is not well documented in the literature. While this could be due to various reasons such as patent restrictions, the lack of information about system design precludes designers who wish to design systems for new languages, for example, from learning from the design procedures employed.

Note

1. The term "graphic symbol" is used in this article to describe two-dimensional representations that take the form of pictures or line drawings [37,96]. This definition therefore encompasses a wide range of more or less linguistic representations but excludes traditional orthography. For a detailed classification of aided symbols used in the field of AAC, see Pampoulou and Fuller [37].

Acknowledgements

Many thanks to the research assistants who helped with data extraction

Funding

This scoping review was made possible with the support from the South African Centre for Digital Language Resources (SADiLaR). SADiLaR is a research infrastructure established the Department

of Science and Technology of the South African government as part of the South African Research Infrastructure Roadmap(SARIR).

Disclosure statement

No potential conflict of interest was reported by the author(s).

ORCID

Kerstin M. Tönsing  <http://orcid.org/0000-0003-1317-0474>
 Refilwe E. Morwane  <http://orcid.org/0000-0001-8881-2297>
 Annalu Waller  <http://orcid.org/0000-0002-3009-8040>

References

- [1] Lynch Y, McCleary M, Smith M. Instructional strategies used in direct AAC interventions with children to support graphic symbol learning: a systematic review. *Child Lang Teach Ther.* 2018;34(1):23–36.
- [2] Russo MJ, Prodan V, Meda NN, et al. High-technology augmentative communication for adults with post-stroke aphasia: a systematic review. *Expert Rev Med Devices.* 2017;14(5):355–370.
- [3] Crowe B, Machalicek W, Wei Q, et al. Augmentative and alternative communication for children with intellectual and developmental disability: a mega-review of the literature. *J Dev Phys Disabil.* 2022;34(1):1–42.
- [4] Andzik NR, Chung YC. Augmentative and alternative communication for adults with complex communication needs: a review of single-case research. *Commun Disord Q.* 2022;43(3):182–194.
- [5] Beukelman DR, Light JC. Augmentative and alternative communication: supporting children and adults with complex communication needs, 5th ed. London: Brookes; 2020.
- [6] Tenny C. A linguist looks at AAC: language representation systems for augmentative and alternative communication, compared with writing systems and natural language. *Writ Syst Res.* 2016;8(1):84–119.
- [7] Fuller DR, Pampoulou E. Opinion: revisiting the means to select and transmit of the AAC model. *JET.* 2022;16(1):321–339.
- [8] Tönsing K, Bartram, J Morwane, R, et. al. Designing electronic graphic symbol-based AAC systems: a scoping review. Part 2: application of human-centred design. *Disabil Rehabil Assist Technol.* 2022 [cited 2022];
- [9] Zangari C, Lloyd L, Vicker B. Augmentative and alternative communication: an historic perspective. *Augment Altern Commun.* 1994;10(1):27–59.
- [10] Shane H, Laubscher E, Schlosser RW, et al. Enhancing communication for individuals with autism: a guide to the visual immersion system. London: Brookes; 2015.
- [11] Smith M. Speech, language and aided communication: connections and questions in a developmental context. *Disabil Rehabil.* 2006;28(3):151–157.
- [12] Kent-Walsh J, Murza KA, Malani MD, et al. Effects of communication partner instruction on the communication of individuals using AAC: a meta-analysis. *Augment Altern Commun.* 2015;31(4):271–284.
- [13] Sennott SC, Light JC, McNaughton D. AAC modeling intervention research review. *Res Pract Pers with Sev Disabil.* 2016;41(2):101–115.
- [14] Moorcroft A, Scarinci N, Meyer C. Speech pathologist perspectives on the acceptance versus rejection or abandonment of AAC systems for children with complex communication needs. *Augment Altern Commun.* 2019;35(3):193–204.
- [15] Johnson JM, Inglebret E, Jones C, et al. Perspectives of speech language pathologists regarding success versus abandonment of AAC. *Augment Alternat Commun.* 2006;22(2):85–99.
- [16] Martin JK, Martin LG, Stumbo NJ, et al. The impact of consumer involvement on satisfaction with and use of assistive technology. *Disabil Rehabil Assist Technol.* 2011;6(3):225–242.
- [17] Moorcroft A, Scarinci N, Meyer C. A systematic review of the barriers and facilitators to the provision and use of low-tech and unaided AAC systems for people with complex communication needs and their families. *Disabil Rehabil Assist Technol.* 2019;14(7):710–731.
- [18] Moorcroft A, Scarinci N, Meyer C. "I've had a love-hate, I mean mostly hate relationship with these PODD books": parent perceptions of how they and their child contributed to AAC rejection and abandonment. *Disabil Rehabil Assist Technol.* 2021;16(1):72–82.
- [19] Pampoulou E. Graphic symbols terminology: a call for a consensus. *JET.* 2017;11(3):92–100.
- [20] Tosi F. Design for ergonomics. NV-1 online. Cham: Springer International Publishing; 2020. (Springer Series in Design and Innovation vol. 2).
- [21] Allen J. Designing desirability in an augmentative and alternative communication device. *Univ Access Inf Soc.* 2005;4(2):135–145.
- [22] Bean A, Cargill LP, Lyle S. Framework for selecting vocabulary for preliterate children who use augmentative and alternative communication. *Am J Speech Lang Pathol.* 2019;28(3):1000–1009.
- [23] Trembath D, Balandin S, Dark L. Why any old words won't do: the importance of vocabulary selection. *Acquir Knowl Speech, Lang Hear.* 2006;8(3):117–119.
- [24] Balandin S, Iacono T. A few well-chosen words. *Augment Altern Commun.* 1998;14(3):147–161.
- [25] Fenson L, Thal D, Reznick JS, et al. MacArthur bates communicative development inventories. London: Brookes; 2007.
- [26] Soto G, Cooper B. An early Spanish vocabulary for children who use AAC: developmental and linguistic considerations. *AAC. Augment Alternat Commun.* 2021;37(1):64–74.
- [27] Schlosser RW, Shane HC, Allen AA, et al. Just-in-time supports in augmentative and alternative communication. *J Dev Phys Disabil.* 2016;28(1):177–193.
- [28] Drager KDR, Light J, Currall J, et al. AAC technologies with visual scene displays and "just in time" programming and symbolic communication turns expressed by students with severe disability. *J Intellect Dev Disabil.* 2019;44(3):321–336.
- [29] Besio S, Chinato MG. A semiotic analysis of the possibilities and limits of Blissymbols. In: Von Tetzchner S, Jensen MH, editors. *European perspectives on augmentative and alternative communication.* London: Whurr; 1996. p. 182–194.
- [30] Fuller DR, Lloyd LL, Stratton MM. Aided AAC symbols. In: Lloyd LL, Fuller DR, Arvidson HH, editors. *Augmentative and alternative communication: a handbook of principles and practices.* Needham Heights (MA): Allyn & Bacon; 1997. p. 48–79.

- [31] Draffan EA, Wald M, Halabi N, et al. Generating acceptable Arabic core vocabularies and symbols for AAC users. In: 6th workshop on speech and language processing for assistive technologies, Dresden, Germany, 2015-08-31, 2015. p. 91–96. <https://eprints.soton.ac.uk/384316/>
- [32] Karal Y, Karal H, Silbir L, et al. Standardization of a graphic symbol system as an alternative communication tool for Turkish. *Educ Technol Soc*. 2016;19(1):53–66.
- [33] Dada S, Huguet A, Bornman J. The iconicity of picture communication symbols for children with English additional language and mild intellectual disability. *Augment Alternat Commun*. 2013;29(4):360–373.
- [34] DeKlerk HM, Dada S, Alant E. Children's identification of graphic symbols representing four basic emotions: comparison of Afrikaans-speaking and Sepedi-speaking children. *J Commun Disord*. 2014;52:1–15.
- [35] Nigam R. Sociocultural development and validation of lexicon for AsianIndian individuals who use augmentative and alternative communication. *Disabil Rehabil Assist Technol*. 2006;1(4):245–256.
- [36] Huer MB. Examining perceptions of graphic symbols across cultures: preliminary study of the impact of culture/ethnicity. *Augment Alternat Commun*. 2000;16(3):180–185.
- [37] Pampoulou E, Fuller DR. Introduction of a new AAC symbol classification system: the multidimensional quaternary symbol continuum (MQSC). *JET*. 2021;15(4):252–267.
- [38] Beukelman DR, Hux K, Dietz A, et al. Using visual scene displays as communication support options for people with chronic, severe aphasia: a summary of AAC research and future research directions. *Augment Alternat Commun*. 2015;31(3):234–245.
- [39] Wilkinson KM, Light J, Drager K. Considerations for the composition of visual scene displays: potential contributions of information from visual and cognitive sciences. *Augment Alternat Commun*. 2012;28(3):137–147.
- [40] Thistle JJ, Wilkinson KM. Building evidence-based practice in AAC display design for young children: current practices and future directions. *Augment Alternat Commun*. 2015;31(2):124–136.
- [41] Wilkinson KM, Jagaroo V. Contributions of principles of visual cognitive science to AAC system display design. *AAC Augment Alternat Commun*. 2004;20(3):123–136.
- [42] Fallon KA, Light J, Achenbach A. The semantic organization patterns of young children: implications for augmentative and alternative communication. *AAC Augment Alternat Commun*. 2003;19(2):74–85.
- [43] Light J, Drager K, McCarthy J, et al. Performance of typically developing four- and five-year-old children with AAC systems using different language organization techniques. *Augment Alternat Commun*. 2004;20(2):63–88.
- [44] Dukhovny E, Gahl S. Manual motor-plan similarity affects lexical recall on a speech-generating device: implications for AAC users. *J Commun Disord*. 2014;48(1):52–60.
- [45] Bedwani MAN, Bruck S, Costley D. Augmentative and alternative communication for children with autism spectrum disorder: an evidence-based evaluation of the language acquisition through motor planning (LAMP) programme. *Cogent Educ*. 2015;2(1):1045807.
- [46] Thistle JJ, Holmes SA, Horn MM, et al. Consistent symbol location affects motor learning in preschoolers without disabilities: implications for designing augmentative and alternative communication displays. *Am J Speech Lang Pathol*. 2018;27(3):1010–1017.
- [47] Pullin G, Hennig S. 17 Ways to say yes: toward nuanced tone of voice in AAC and speech technology. *AAC Augment Alternat Commun*. 2015;31(2):170–180.
- [48] Schlünz GI, Gumede T, Wilken I, et al. Applications in accessibility of text-to-speech synthesis for South African languages: initial system integration and user engagement. In *ACM International Conference Proceeding Series*. 2017.
- [49] Nekoto W, Marivate V, Matsila T, et al. Participatory research for low-resourced machine translation: a case study in African languages. In: *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*. Stroudsburg (PA): Association for Computational Linguistics; 2020. p. 2144–60. <https://arxiv.org/pdf/2010.02353>
- [50] McNaughton D, Light J. The iPad and mobile technology revolution: benefits and challenges for individuals who require augmentative and alternative communication. *Augment Alternat Commun*. 2013;29(2):107–116.
- [51] Ogletree BT, McMurry S, Schmidt M, et al. The changing world of augmentative and alternative communication (AAC): examining three realities faced by today's AAC provider. *Perspect ASHA SIGs*. 2018;3(12):113–122.
- [52] Dada S, Murphy Y, Tönsing K. Augmentative and alternative communication practices: a descriptive study of the perceptions of South African speech-language therapists. *Augment Alternat Commun*. 2017;33(4):189–200.
- [53] Boster JB, McCarthy JW. Designing augmentative and alternative communication applications: the results of focus groups with speech-language pathologists and parents of children with autism spectrum disorder. *Disabil Rehabil Assist Technol*. 2018;13(4):353–365.
- [54] Frick B, Boster JB, Thompson S. Animation in AAC: previous research, a sample of current availability in the United States, and future research potential. *Assist Technol*. 2022 [cited 15 Nov 2022];1–10. doi:10.1080/10400435.2022.2043954
- [55] de Couvreur L, Goossens R. Design for (every)one: co-creation as a bridge between universal design and rehabilitation engineering. *CoDesign*. 2011;7(2):107–121.
- [56] Amery R, Thirumanickam A, Barker R, et al. Developing augmentative and alternative communication systems in languages other than English: a scoping review. *Am J Speech Lang Pathol*. 2022 [cited 15 Nov 2022]; 1–20. doi:10.1044/2022_ASHA-21-00396
- [57] Light J, McNaughton D, Beukelman D, et al. Challenges and opportunities in augmentative and alternative communication: research and technology development to enhance communication and participation for individuals with complex communication needs. *Augment Alternat Commun*. 2019;35(1):1–12.
- [58] Sucharew H, Macaluso M. Methods for research evidence synthesis: the scoping review approach. *J Hosp Med*. 2019;14(7):416–418.
- [59] Arksey H, O'Malley L. Scoping studies: towards a methodological framework. *Int J Soc Res Methodol Theory Pract*. 2005;8(1):19–32.
- [60] Levac D, Colquhoun H, O'Brien KR. Scoping studies: advancing the methodology. *Implement Sci*. 2010;5(69):1–18.

- [61] Tricco AC, Lillie E, Zarin W, et al. PRISMA extension for scoping reviews (PRISMA-ScR): checklist and explanation. *Ann Intern Med*. 2018;169(7):467–473.
- [62] Kellermeyer L, Harnke B, Knight S, et al. The impact of patient, intervention, comparison, outcome (PICO) as a search strategy tool on literature search quality: a systematic review. *J Med Libr Assoc*. 2018;106(4):203–206.
- [63] World Bank Group. World Bank country and lending groups: country classification [Internet]. 2022. <https://data-helpdesk.worldbank.org/knowledgebase/articles/906519-world-bank-country-and-lending-groups>
- [64] Hill K. Augmentative and alternative communication (AAC) research and development: the challenge of evidence-based practice. *Int J Comput Process Orient Lang*. 2006;19(4):1–14.
- [65] Bhattacharya S, Basu A. Design of an iconic communication aid for individuals in India with speech and motion impairments. *Assist Technol*. 2009;21(4):173–187.
- [66] Al-Arifi B, Al-Rubaian A, Al-Ofisan G, et al. Towards an Arabic language augmentative and alternative communication application for autism. In Marcus A, editor. *Design, user experience, and usability health, learning, playing, cultural, and cross-cultural user experience*. Berlin, Heidelberg: Springer; 2013. p. 333–341.
- [67] Cheung KLKH, Lam THW, Cheung KLKH. A mobile augmentative and alternative communication (MAAC) application for disabilities. In: *Heal 2014 – 7th Int Conf Heal Informatics, Proceedings; Part 7th Int Jt Conf Biomed Eng Syst Technol BIOSTEC 2014*. 2014. p. 188–95.
- [68] Karita T, Faculty of Education, Ehime University. Development of a communication aid app with iOS devices to support children/persons with speech disabilities. *J Adv Comput Intell Inform*. 2017;21(2):371–377.
- [69] Rodríguez-Sedano F, Conde-González MA, Fernández-Llamas C, et al. The use of a new visual language as a supporting resource for people with intellectual disabilities. In: Zaphiris P, Ioannou A, editors. *Learning and collaboration technologies technology in education*. Cham: Springer International Publishing; 2017. p. 202–214 (Lecture Notes in Computer Science; vol. 10296).
- [70] Hiroto T. An AAC system designed for improving behaviors and attitudes in communication between children with CCN and their peers. In: *Lecture notes in computer science (including subseries lecture notes in artificial intelligence and lecture notes in bioinformatics)*. New York, NY: Springer International Publishing; 2018. p. 530. 41.
- [71] Mahmud AA, Limpens Y, Martens J-B. Expressing through digital photographs: an assistive tool for persons with aphasia. *Univ Access Inf Soc*. 2013;12(3):309–326.
- [72] van de Sandt-Koenderman M, Wiegiers J, Hardy P. A computerised communication aid for people with aphasia. *Disabil Rehabil*. 2005;27(9):529–533.
- [73] Daemen E, Dadlani P, Du J, et al. Designing a free style, indirect, and interactive storytelling application for people with aphasia. In: *Lecture notes in computer science (including subseries lecture notes in artificial intelligence and lecture notes in bioinformatics)*. 2007. p. 221–234.
- [74] de Oliveira K, Junior J, Silva J, et al. A. VoxLaps: a free symbol-based AAC application for Brazilian Portuguese. In: Antona M, Stephanidis C, editors. *Universal access in human-computer interaction*. Toronto, Canada: Springer; 2016. p. 129–140.
- [75] Hervás R, Bautista S, Méndez G, et al. Predictive composition of pictogram messages for users with autism. *J Ambient Intell Hum Comput*. 2020;11(11):5649–5664.
- [76] Martin E, Cupeiro C, Pizarro L, et al. “Today I tell” a comics and story creation app for people with autism spectrum condition. *Int J Hum-Comput Interact*. 2019;35(8):679–691.
- [77] Williams K, Moffatt K, McCall D, et al. Designing conversation cues on a head-worn display to support persons with aphasia. In: *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*. New York, NY, USA: ACM; 2015. p. 231–40.
- [78] da Silva DP, Amate FC, Basile FRM, et al. AACVOX: mobile application for augmentative alternative communication to help people with speech disorder and motor impairment. *Res Biomed Eng*. 2018;34(2):166–175.
- [79] Bruno J. Gateway for Snap: user manual [Internet]. Communication Technology Resources; 2020. http://www.gatewaytolanguageandlearning.com/resources/ewExternalFiles/GatewayTeen_AdultFunctionalManual.pdf
- [80] Porter G, Cafiero JM. Pragmatic organization dynamic display (PODD) communication books: a promising practice for individuals with autism spectrum disorders. *Perspect Augment Alternat Commun*. 2009;18(4):121–129.
- [81] Judge S, Randall N, Goldbart J, et al. The language and communication attributes of graphic symbol communication aids – a systematic review and narrative synthesis. *Disabil Rehabil Assist Technol*. 2019;3107:1–11.
- [82] Light J, McNaughton D, Caron J. New and emerging AAC technology supports for children with complex communication needs and their communication partners: state of the science and future research directions. *Augment Alternat Commun*. 2019;35(1):26–41.
- [83] Waller A. Telling tales: unlocking the potential of AAC technologies. *Int J Lang Commun Disord*. 2019;54(2): 159–169.
- [84] Valencia S, Pavel A, Santa Maria J, et al. Conversational agency in augmentative and alternative communication. In: *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. New York, NY, USA: ACM; 2020. p. 1–12.
- [85] Di Mascio T, Tarantino L, Cirelli L, et al. Designing a personalizable ASD-oriented AAC tool: an action research experience. *Adv Intell Syst Comput*. 2019;804:200–209.
- [86] Jafri R, Almasoud AM, Alshammari RMT, et al. A low-cost gaze-based Arabic augmentative and alternative communication system for people with severe speech and motor impairments. In: Stephanidis C, Antona M, editors. *Lecture notes in computer science (including subseries lecture notes in artificial intelligence and lecture notes in bioinformatics)*. New York (NY): Springer International Publishing; 2020. p. 279–290.
- [87] Thistle JJ, Wilkinson K. Effects of background color and symbol arrangement cues on construction of multi-symbol messages by young children without disabilities: implications for aided AAC design. *Augment Alternat Commun*. 2017;33(3):160–169.
- [88] Murray J, Lynch Y, Meredith S, et al. Professionals’ decision-making in recommending communication aids in the UK: competing considerations. *Augment Alternat Commun*. 2019;35(3):167–179.
- [89] Gregor S, Hevner AR, The Australian National University. Positioning and presenting design science research for maximum impact. *MISQ*. 2013;37(2):337–355.

- [90] Wieringa R. Design science methodology: principles and practice. In: *Proceedings – International Conference on Software Engineering*. 2010. p. 493–4.
- [91] World Health Organization. World Report on Disability [Internet]2011. https://www.who.int/disabilities/world_report/2011/report.pdf
- [92] Smith RO, Scherer MJ, Cooper R, et al. Assistive technology products: a position paper from the first global research, innovation, and education on assistive technology (GREAT) summit. *Disabil Rehabil Assist Technol*. 2018; 13(5):473–485.
- [93] Bornman J, Bryen DN, Moolman E, et al. Use of consumer wireless devices by South Africans with severe communication disability. *Afr J Disabil*. 2016;5(1):1–9.
- [94] Tönsing KM, van Niekerk K, Schlünz GI, et al. AAC services for multilingual populations: South African service provider perspectives. *J Commun Disord*. 2018;73:62–76.
- [95] Grosjean F. Bilingualism: a short introduction. In: Grosjean F, Li P, editors. *The psycholinguistics of bilingualism*. Chichester: Wiley-Blackwell; 2013. p. 5–25.
- [96] Loncke F. *Augmentative and alternative communication: models and applications*. 2nd ed. San Diego (CA): Plural Publishing; 2022.
- [97] An S, Feng X, Dai Y, et al. Development and evaluation of a speech-generating AAC mobile app for minimally verbal children with autism spectrum disorder in Mainland China. *Mol Autism*. 2017;8(1):1–12.
- [98] Babic J, Slivar I, Car Z, et al. Prototype-driven software development process for augmentative and alternative communication applications. In: *2015 13th International Conference on Telecommunications (ConTEL)*. New York, NY: IEEE; 2015. p. 1–8.
- [99] Boyd-Graber J, Nikolova S, Moffatt K, et al. Participatory design with proxies: developing a desktop-PDA system to support people with aphasia. *Conf Hum Factors Comput Syst – Proc*, 2006, vol. 1. p. 151–160.
- [100] de Faria Borges LCL, Filgueiras LVL, Maciel C, et al. The life cycle of a customized communication device for a child with cerebral palsy: contributions toward the PD4CAT method. *J Brazilian Comput Soc*. 2014;20(1):10.
- [101] Hayes GR, Hirano S, Marcu G, et al. Interactive visual supports for children with autism. *Pers Ubiquit Comput*. 2010;14(7):663–680.
- [102] Hine N, Arnott JL, Smith D. Design issues encountered in the development of a mobile multimedia augmentative communication service. *Univ Access Inf Soc*. 2003;2(3): 255–264.
- [103] Lubas M, Mitchell J, De Leo G. User-centered design and augmentative and alternative communication apps for children with autism spectrum disorders. *SAGE Open*. 2014;4(2):215824401453750.
- [104] Mendes M, Correia S. Combining research, theory and end user experiments for suitable AAC apps. *Assist Technol Res Ser*. 2013;33:340–346.
- [105] Saturno CE, Ramirez ARG, Conte MJ, et al. An augmentative and alternative communication tool for children and adolescents with cerebral palsy. *Behav Inf Technol*. 2015; 34(6):632–645.
- [106] Stancić Z, Škrinjar JJF, Car Ž, et al. Systems of support for persons with complex communication needs. In: *2013 36th International Convention on Information and Communication Technology, Electronics and Microelectronics (MIPRO)*. 2013. p. 700–4.

Copyright of Disability & Rehabilitation: Assistive Technology is the property of Taylor & Francis Ltd and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.